Online Resource 1: Supplementary methods and results

Historical abstract loss and use-dependent robustness of a semantic novelty indicator

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Supplementary information for the author preprint. Original frozen results, earlier post-confirmation checks and separately labelled post hoc descriptive analyses. Online Resource 2 contains the full numerical CSV/JSON tables. Reproducibility documentation records acquisition, computational verification and execution provenance separately.

S1 Cohort construction and complete accounting

Both studies compare a fixed 2021 focal cohort with records published in 2018–2020. They use selected collections in which the observed title and abstract are available, so that a controlled loss of abstract information can be measured against the same papers' observed baseline. Completeness here concerns this selected current metadata snapshot, including retraction or concern flags that can postdate publication; it does not establish what a database displayed when a paper was published.

The ACL/EMNLP frame comprises eligible main-conference long and short research papers. Findings, workshops, demonstrations, tutorials, student sessions and front matter are outside the declared frame. Venue names, including the joint-event naming of ACL 2021, define source groups rather than a validated disciplinary classification. Anthology XML lacks a validated record-level language classification; English venue context is not individual language verification. The historical frame contains 3,799 records after removing one exact title-plus-abstract duplicate. The distinct IDs D18-1092 and D18-1105 also have distinct DOIs, but their pagination-normalized extracted PDF texts were identical. The first record in the deterministic ordering was retained. There were no further duplicate or text-completeness exclusions among the 1,557 focal records. Deduplication used identifiers, normalized DOIs and exact normalized text, not an exhaustive fuzzy version audit.

The biomedical frame began with 4,896 publisher-classified full Research Articles from PLOS Computational Biology (PLOSCB) and PLOS Genetics (PLOSGEN). An exact normalized DOI join connected each candidate to PubMed. Eligibility required a unique, valid full electronic publication date whose year agreed with the publisher date; issue and indexing dates were not fallbacks. All candidates joined uniquely, with no date, language, title or abstract losses and no exact-text duplicates. A tested date-free DOI-rescue branch was not needed on the acquired data.

Thirty-four distinct candidates were excluded: 24 solely by publication type, six by both type and retraction relationship, and four solely by expression-of-concern relationship. The type flags comprise 19 Meta-Analysis, three Review, two Systematic Review and six Retracted Publication records. The ten relationship flags comprise the same six retractions and four expressions of concern. Thus the 30 type flags and ten relationship flags overlap; they are not 40 excluded papers. Ordinary linked errata did not automatically exclude the original article. Current retraction or concern flags may postdate publication.

The biomedical inclusion gate required at least 200 focal records per journal, at least 2,000 historical records, and at least 95% joint text availability under both the valid-frame and publisher-frame denominators in every journal-year. Retained counts were 3,391 history and 1,471 focal records. Publisher-frame coverage ranged from 98.168% to 100%; valid-frame availability was 100%. These denominators distinguish eligibility exclusions from unavailable text.

Table S1: Cohort accounting. Panel A gives retained records; 2018–2020 form history and 2021 is focal. Panel B gives biomedical publisher-frame accounting. The valid frame equals the retained count because every record eligible after date, type, identifier and retraction/concern exclusions has both texts. Coverage divides retained records by publisher candidates; text availability within each valid frame is 100%.

A. Retained records

Year	ACL	EMNLP	PLOSCB	PLOSGEN
2018	381	548	528	536
2019	660	681	620	496
2020	778	751	706	505
2021	710	847	934	537
History total	3,799		3,391	
Focal total	1,557		1,471	

B. Biomedical publisher frame

Year	Source	Candidates	Excluded	Retained	Coverage (%)
2018	PLOSCB	528	0	528	100.000
2018	PLOSGEN	546	10	536	98.168
2019	PLOSCB	622	2	620	99.678
2019	PLOSGEN	501	5	496	99.002
2020	PLOSCB	710	4	706	99.437
2020	PLOSGEN	513	8	505	98.441
2021	PLOSCB	936	2	934	99.786
2021	PLOSGEN	540	3	537	99.444
Total		4,896	34	4,862	

S2 Study sequence and protocol specifications

The exploratory pilot used 2,270 historical papers from 2018–2019, a fixed-seed sample of 300 focal papers from 2020, and 20 masks per condition. It informed the later emphasis on aggregate, ranking and selection stability and the precision planning for 200 masks. That emphasis was not the original pilot hypothesis. Confirmation then included all eligible 2021 ACL/EMNLP focal records.

Both structured pilot controls matched historical year. Before confirmation, the lexical-proxy control was changed to year × source matching; the source-preferred control remained year-matched. The cutoff diagnostics, operating-threshold grid and 45-mean inference family were also specified for confirmation. These changes were prospective relative to confirmation outcomes, not retroactive changes to the pilot.

The biomedical design followed the ACL findings but preceded biomedical outcomes. It transported random and source-preferred masking at 25% and 50%, with PLOSCB fixed as the preferred source. It did not transport the ACL lexical proxy. The collection was acquired separately, but the encoder, implementation, operational thresholds and author were shared. The two studies therefore remain separate analyses, not independent replications of every design choice.

The ACL confirmation was locally frozen on 4 October 2026 at 20:19:05 UTC; biomedical replication was locally frozen on 5 October 2026 at 09:25:32 UTC. These documented local freezes are not externally registered preregistrations. The later OpenAlex audit in Section S10 was undertaken after both studies; it does not alter their original protocols or calibrate their already selected rates.

S3 Representations, title substitution and matched controls

The baseline score is the existing nearest-prior-paper semantic distance,

$$b_i = 1 - \max_{j \in H} \langle z_i, z_j \rangle,$$

where H contains strictly earlier eligible papers and vectors are normalized. Similarities are exact. The primary score uses the maximum similarity, not a neighborhood average. The pilot diagnostic separately uses 20 title neighbors (Section S7). The pinned SPECTER encoder uses final-layer CLS pooling, L2 normalization, float32 and a 512-token cap. The input separator is the tokenizer separator token (`[SEP]`). Full input is title, separator and abstract; title-only input is the authentic title followed by the same separator. "Full" denotes this text input, not the full article. Strict temporal ordering of comparison papers does not constrain the pretrained encoder's training chronology.

Historical title substitution replaces selected historical title-plus-abstract representations with their authentic title-only counterparts while keeping every focal full representation fixed. Other recorded policies remove masked history, use titles on both sides, or restore full representations. Their interpretations differ. Restoration equality and all-title mask invariance are implementation checks, not validation of scientific originality or evidence that title-only and full scoring are interchangeable. Removing historical candidates cannot decrease nearest-neighbor distance while the retained history is nonempty. This property does not apply to title substitution.

ACL inputs had no full or title-only truncation. Biomedical full inputs exceeded the token limit for 67 of 4,862 papers (1.378%): 22 PLOSCB and 45 PLOSGEN papers across history and focal years together. No title-only inputs were truncated and no papers were excluded for length. The effect of this source-related truncation on the differing mask responses was not isolated.

For rate p and history size h, each assignment masks exactly $c = \lfloor hp + 0.5 \rfloor$ records: 380, 950 and 1,900 in ACL at 10%, 25% and 50%, and 848 and 1,696 in biomedical history at 25% and 50%. Uniform assignments sample without replacement. For structured assignments, draw independent exponential clocks with rate 8 for preferred records and rate 1 for others, then select the c smallest clocks. These rates specify a sampling preference, not an exact 8:1 ratio of marginal inclusion probabilities. ACL preference targets the ACL source or titles containing the case-insensitive substring `translat`; biomedical preference targets PLOSCB. This lexical rule is an outcome-independent proxy, not a validated scientific topic.

For each structured assignment, the independently seeded matched control samples uniformly within cells, preserving its realized missing count in every cell. Source-preferred controls match historical year; ACL lexical-proxy controls match historical year $\times$ source. Zero and fully masked cells are retained. Matching the preferred source or lexical stratum itself would remove the intended intervention. The controls fix temporal opportunity, and source composition for the lexical comparison, while leaving other content and neighborhood differences unmatched. They do not identify an isolated causal effect of a source label.

The same historical title-substitution assignment is applied to every focal paper. The assignment, not a paper, is the intervention unit. Each structured assignment and its matched control form one paired unit. ACL contains 1,800 base assignments and 1,200 matched controls; biomedical contains 800 and 400, respectively. Every condition has exactly 200 replicates, without adaptive extension. Rates are synthetic scenarios, including a 50% stress condition; neither lower rate nor the structured preference is an estimated real-world omission mechanism.

The original seed masters are 3216741901 for ACL/EMNLP and 2785196430 for the biomedical experiment; the namespaces are `acl-2021-confirmation-v1` and `pubmed-2021-replication-v1`. Assignment identifiers encode study, mechanism, rate in millionths, and a zero-based replicate index. For each identifier and role (mask or control), the compact JSON array [namespace, master seed, assignment identifier, role] is hashed with SHA-256. Its first 16 bytes are split into four big-endian unsigned 32-bit integers and passed to NumPy `SeedSequence`, then `default_rng` (PCG64). Mask and control seeds are distinct. Complete seed vectors and ordered record identifiers are provided with the reproducibility materials; changing record order can change an assignment even with the same seed.

Table S2: Assignment counts and matching-cell degeneracy.

Condition	Masked	Base n	Matching	Zero	Full
ACL					
Random 10%	380	200	–	–	–
Source-preferred 10%	380	200	Year	0	0
Lexical-proxy 10%	380	200	Year × source	0	0
Random 25%	950	200	–	–	–
Source-preferred 25%	950	200	Year	0	0
Lexical-proxy 25%	950	200	Year × source	0	0
Random 50%	1900	200	–	–	–
Source-preferred 50%	1900	200	Year	0	0
Lexical-proxy 50%	1900	200	Year × source	0	0
Biomedical					
Random 25%	848	200	–	–	–
Source-preferred 25%	848	200	Year	0	0
Random 50%	1696	200	–	–	–
Source-preferred 50%	1696	200	Year	0	0

Zero/full columns sum the respective cell instances across 200 controls, not distinct papers. Each structured condition has 200 matched controls; random conditions have none. SD/quantiles over masks are supplied in the complete numerical CSV files.

S4 Endpoints and uncertainty

Let s_0 be the population SD of all focal baseline scores, Δ_0 the baseline difference between source means, and Δ_m the same contrast under assignment m. The aggregate endpoint is $G_m = |\Delta_m - \Delta_0|/s_0$, taking the absolute value before averaging over masks. The reference contrasts are ACL minus EMNLP and PLOSCB minus PLOSGEN. Their baseline differences are $-0.130701s_0$ and $+0.261287s_0$. A small standardized change can still be appreciable relative to a small original contrast.

Let S_0 and S_m contain the highest-scoring $k = \lceil 0.10n \rceil$ focal papers at baseline and under masking. The selection endpoint is $T_m = |S_0 \setminus S_m|/k$, with $k = 156$ for ACL and 148 for biomedical. It is a fraction of the original selected set, not all focal papers. Lexical record IDs resolve exact score ties for membership. Rank displacement is $Q_m = n^{-1} \sum_i |r_{im} - r_{i0}|/(n-1)$; exact score ties receive average ranks. No assignment changes the focal cohort.

The operational tolerances are $G = 0.10$, $T = 0.05$ and $Q = 0.01$. They have different units and are not externally validated policy thresholds. The primary coexistence rule is $U_G < 0.10$ and $L_T > 0.05$, where U and L are simultaneous mean bounds. Touching a boundary remains unresolved. A lower Q bound above 0.01 is additional evidence and cannot replace a failed selection criterion. The selection threshold represents 7.8 or 7.4 expected lost selections; the rank threshold represents mean movements of 15.56 or 14.70 rank positions.

Each mean interval is

$$\overline{x} \pm t_{199,\,1-0.05/(2J)} \frac{s_x}{\sqrt{200}}.$$

ACL uses $J = 45$: 27 mixed-representation means and 18 paired means. Biomedical uses $J = 18$: 12 mixed means and six paired means. The respective critical values are 3.3090595211511644 and 3.029133739375753. Paired outcomes subtract the matched control within each replicate before computing SD and Monte Carlo SE (MCSE). Intervals are untruncated, and incomplete or nonfinite primary replicates are rejected rather than silently omitted. Bonferroni does not require independence among endpoints, but Student coverage is approximate for bounded, potentially skewed outcomes. These are two separate approximate 95% families; there is no joint 95% claim across both studies or formal collection-interaction test.

The intervals describe uncertainty in expected effects under the fixed assignment generator, conditional on the corpus and encoder. SD and distribution quantiles describe variation between assignments; MCSE describes uncertainty in their mean. A pointwise Wilson interval for the descriptive fraction of masks jointly satisfying

$G < 0.10$ and $T > 0.05$ is outside the primary family. Mean-level coexistence need not hold in most individual masks.

S5 Full primary and paired results

Tables S3 and S4 report all primary conditions and structured-control comparisons, including unfavorable and unresolved effects. Primary means retain their separate study families; paired signs refer to the structured assignment minus its matched control.

Table S3: All original primary means and distribution/precision summaries.

Condition	Metric	Mean	Mask SD	MCSE	Interval
ACL					
Random 10%	G	0.0088	0.0068	0.00048	[0.0072, 0.0104]
Random 10%	T	6.25	1.78	0.126	[5.83, 6.66]
Random 10%	Q	2.23	0.27	0.019	[2.17, 2.30]
Source-preferred 10%	G	0.0069	0.0055	0.00039	[0.0056, 0.0082]
Source-preferred 10%	T	5.39	1.62	0.115	[5.0112, 5.77]
Source-preferred 10%	Q	2.04	0.26	0.018	[1.98, 2.10]
Lexical-proxy 10%	G	0.0114	0.0073	0.00051	[0.0097, 0.0131]
Lexical-proxy 10%	T	4.59	1.54	0.109	[4.23, 4.95]
Lexical-proxy 10%	Q	2.31	0.29	0.020	[2.24, 2.38]
Random 25%	G	0.0175	0.0103	0.00073	[0.0151, 0.0199]
Random 25%	T	14.10	2.28	0.161	[13.56, 14.63]
Random 25%	Q	5.18	0.35	0.025	[5.09, 5.26]
Source-preferred 25%	G	0.0129	0.0095	0.00067	[0.0106, 0.0151]
Source-preferred 25%	T	12.59	1.87	0.132	[12.15, 13.03]
Source-preferred 25%	Q	4.91	0.35	0.025	[4.83, 4.99]
Lexical-proxy 25%	G	0.0305	0.0138	0.00097	[0.0273, 0.0338]
Lexical-proxy 25%	T	13.35	2.27	0.160	[12.82, 13.88]
Lexical-proxy 25%	Q	6.04	0.35	0.025	[5.95, 6.12]
Random 50%	G	0.0352	0.0191	0.00135	[0.0308, 0.0397]
Random 50%	T	26.92	2.15	0.152	[26.41, 27.42]
Random 50%	Q	9.60	0.37	0.026	[9.51, 9.68]
Source-preferred 50%	G	0.0277	0.0159	0.00112	[0.0240, 0.0314]
Source-preferred 50%	T	24.25	1.57	0.111	[23.88, 24.62]
Source-preferred 50%	Q	9.25	0.29	0.021	[9.18, 9.32]
Lexical-proxy 50%	G	0.0471	0.0201	0.00142	[0.0424, 0.0518]
Lexical-proxy 50%	T	27.22	2.30	0.163	[26.69, 27.76]
Lexical-proxy 50%	Q	11.00	0.34	0.024	[10.92, 11.08]
Biomedical					
Random 25%	G	0.0167	0.0128	0.00090	[0.0139, 0.0194]
Random 25%	T	14.90	2.27	0.160	[14.42, 15.39]
Random 25%	Q	5.41	0.34	0.024	[5.34, 5.48]
Source-preferred 25%	G	0.1486	0.0159	0.00113	[0.1452, 0.1520]
Source-preferred 25%	T	17.91	2.13	0.150	[17.45, 18.36]
Source-preferred 25%	Q	6.02	0.34	0.024	[5.95, 6.09]
Random 50%	G	0.0267	0.0208	0.00147	[0.0223, 0.0311]
Random 50%	T	25.55	2.40	0.170	[25.03, 26.06]
Random 50%	Q	9.79	0.36	0.026	[9.71, 9.87]
Source-preferred 50%	G	0.3770	0.0261	0.00185	[0.3715, 0.3826]
Source-preferred 50%	T	32.03	2.06	0.146	[31.59, 32.48]
Source-preferred 50%	Q	11.28	0.37	0.026	[11.20, 11.35]

G uses baseline SD; T, Q are percentages of selected-set size/rank range. Intervals belong to separate approximate 45/18-mean families. MCSE is mask SD divided by $\sqrt{200}$.

Matched-control means (descriptive).

Condition	G	T (%)	Q (%)
ACL			
Source-preferred 10%	0.0083	6.15	2.28
Lexical-proxy 10%	0.0091	6.02	2.21
Source-preferred 25%	0.0170	14.90	5.33
Lexical-proxy 25%	0.0155	14.14	5.20
Source-preferred 50%	0.0343	27.33	9.71
Lexical-proxy 50%	0.0323	27.37	9.54
Biomedical			
Source-preferred 25%	0.0171	15.02	5.43
Source-preferred 50%	0.0285	25.81	9.80

Table S4: All original paired structured-minus-control effects.

Condition	Metric	Mean	Pair SD	MCSE	Interval
ACL					
Source-preferred 10%	G	-0.00146	0.00790	0.000559	[-0.00331, 0.00039]
Source-preferred 10%	T	-0.760	2.374	0.1679	[-1.315, -0.204]
Source-preferred 10%	Q	-0.237	0.375	0.0265	[-0.324, -0.149]
Lexical-proxy 10%	G	0.00232	0.01043	0.000738	[-0.00013, 0.00476]
Lexical-proxy 10%	T	-1.429	2.298	0.1625	[-1.967, -0.892]
Lexical-proxy 10%	Q	0.100	0.420	0.0297	[0.002, 0.198]
Source-preferred 25%	G	-0.00416	0.01499	0.001060	[-0.00767, -0.00066]
Source-preferred 25%	T	-2.308	2.956	0.2090	[-2.999, -1.616]
Source-preferred 25%	Q	-0.416	0.495	0.0350	[-0.532, -0.301]
Lexical-proxy 25%	G	0.01506	0.01839	0.001300	[0.01076, 0.01936]
Lexical-proxy 25%	T	-0.792	3.301	0.2334	[-1.564, -0.019]
Lexical-proxy 25%	Q	0.842	0.506	0.0358	[0.724, 0.960]
Source-preferred 50%	G	-0.00662	0.02605	0.001842	[-0.01272, -0.00052]
Source-preferred 50%	T	-3.077	2.811	0.1987	[-3.735, -2.419]
Source-preferred 50%	Q	-0.462	0.485	0.0343	[-0.576, -0.349]
Lexical-proxy 50%	G	0.01475	0.02821	0.001995	[0.00815, 0.02135]
Lexical-proxy 50%	T	-0.141	3.129	0.2213	[-0.873, 0.591]
Lexical-proxy 50%	Q	1.462	0.441	0.0312	[1.359, 1.565]
Biomedical					
Source-preferred 25%	G	0.13154	0.02017	0.001426	[0.12722, 0.13587]
Source-preferred 25%	T	2.889	3.188	0.2254	[2.206, 3.571]
Source-preferred 25%	Q	0.586	0.471	0.0333	[0.485, 0.686]
Source-preferred 50%	G	0.34857	0.03244	0.002294	[0.34163, 0.35552]
Source-preferred 50%	T	6.223	3.082	0.2179	[5.563, 6.883]
Source-preferred 50%	Q	1.477	0.495	0.0350	[1.371, 1.583]

G uses baseline SD; T, Q are percentage points. Source-preferred controls match year in both studies; lexical-proxy controls match year by source. Unresolved signs are retained.

All nine ACL aggregate upper bounds lie below the main tolerance, but the 10% lexical condition does not pass the joint selection criterion. The 10% source condition only narrowly clears that criterion. Biomedical random masking meets both criteria at both rates. Biomedical source-preferred masking instead exceeds the aggregate tolerance while also changing selection. It therefore shows aggregate and selection instability together. These exceptions and differences remain in the complete tables rather than being removed by a pooled summary.

The paired source effects also differ. At 25% and 50%, ACL source preference reduces G, T and Q relative to year-matched controls, whereas biomedical source preference increases them. ACL lexical effects vary by endpoint: G and Q increase at 25%/50%, but T decreases at 10%/25% and is unresolved at 50%. These contrasts do not support universal worsening from structured missingness. The opposite paired directions show that structure has

no uniformly increasing effect in these experiments; their difference is not attributed to discipline, truncation or a particular semantic mechanism.

S6 Selection margins, rank displacement and threshold sensitivity

The baseline cutoff c_0 is the midpoint between the kth and $(k+1)$th scores. A paper's signed margin is $(b_i - c_0)/s_0$. The separate rank neighborhood uses descending ordinal midpoint percentile $(r - 0.5)/n \in [0.09, 0.11]$, including all exact-score ties crossing either edge. ACL's cutoff gap is 0.000896216 SD and its neighborhood contains 31 papers, including 16 original selections. Biomedical's gap is 0.00402522 SD and its neighborhood contains 30 papers, also including 16 original selections. Neither cutoff has exact ties.

Signed-margin bins have boundaries $-0.10, -0.05, -0.025, -0.01, 0, 0.01, 0.025, 0.05, 0.10$ and open-ended tails. Bins are left-closed/right-open, with exact zero in $[0, 0.01)$; empty bins remain reported. For $\delta \in \{0.01, 0.025, 0.05, 0.10\}$, far-margin loss is

$$F_m(\delta) = \frac{\#\{i \in S_0 \setminus S_m : (b_i - c_0)/s_0 > \delta\}}{k}.$$

Both inside- and outside-neighborhood loss fractions use k. Conditional loss among original selections outside the neighborhood instead uses 140 ACL or 132 biomedical papers. Retention and entry probabilities within margin bins use that bin's selected or unselected baseline population. Counts, fractions and denominators must therefore be read together. Rank neighborhoods and positive score margins are distinct, overlapping threshold-based summaries. Let $O = S_0 \setminus B$ contain the baseline selections outside the rank band, and let $C_\delta = \{i \in S_0 : (b_i - c_0)/s_0 > \delta\}$. Under the same descending baseline ordering, both are upper prefixes of S_0, including the stated tie handling, so the sets are nested. Record-level verification at $\delta = 0.10$ gives $C_{0.10} \subset O$ in ACL (132 versus 140 papers), and $O \subset C_{0.10}$ in the biomedical collection (132 versus 133 papers). Intersecting either inclusion with the same lost set $S_0 \setminus S_m$ preserves it. The overlapping counts must not be added. For a verified inclusion, subtracting the smaller lost-subset count from the larger does identify loss from their set difference. The verification changes this interpretation, not any reported mean, interval or selection count. Outside-band counts are constrained by the band's capacity of 16 baseline selections; Section S13.2 gives the bound and conditional-bin loss probabilities.

Table S5: Expected selection losses and all fixed positive score-margin qualifications.

Condition	Total	Outside	Cond.(%)	> .01	> .025	> .05	> .10
ACL							
Random 10%	9.745	1.175	0.84	7.930	4.415	1.920	1.100
Source-preferred 10%	8.410	1.040	0.74	6.440	3.245	1.540	1.020
Lexical-proxy 10%	7.155	1.240	0.89	5.380	2.550	1.405	1.170
Random 25%	21.990	9.135	6.52	20.115	16.770	13.040	4.460
Source-preferred 25%	19.640	8.050	5.75	17.670	14.385	11.210	3.360
Lexical-proxy 25%	20.825	7.675	5.48	19.030	15.505	11.510	4.235
Random 50%	41.990	30.380	21.70	40.295	36.740	33.570	24.495
Source-preferred 50%	37.835	28.065	20.05	35.875	32.730	30.380	21.670
Lexical-proxy 50%	42.470	30.545	21.82	40.865	37.410	34.075	24.370
Biomedical							
Random 25%	22.055	9.040	6.85	19.580	18.580	16.985	9.770
Source-preferred 25%	26.505	13.640	10.33	24.170	23.170	21.335	14.595
Random 50%	37.810	26.865	20.35	36.095	35.095	33.690	27.550
Source-preferred 50%	47.410	35.235	26.69	45.225	44.225	42.630	36.225

Total/outside/margin entries are expected paper counts, not percentages. Outside uses the fixed rank band; Cond. divides outside loss by 140 ACL or 132 biomedical originally selected papers outside the band. Margin columns use baseline SD above the cutoff and overlap the rank-band categories. Divide expected counts by $k = 156$ or 148 for selected-set fractions.

Mean of per-mask movement summaries.

Condition	Median	90th percentile	Fraction > 5 pp
ACL			
Random 10%	1.33	2.02	5.90
Source-preferred 10%	1.19	1.87	5.54
Lexical-proxy 10%	1.40	2.12	6.07
Random 25%	3.69	9.07	20.10
Source-preferred 25%	3.40	8.11	16.35
Lexical-proxy 25%	4.32	10.97	36.67
Random 50%	7.91	18.40	67.30
Source-preferred 50%	7.47	17.85	67.51
Lexical-proxy 50%	9.18	21.36	71.84
Biomedical			
Random 25%	3.77	10.35	19.51
Source-preferred 25%	4.32	11.56	34.12
Random 50%	8.19	19.73	69.43
Source-preferred 50%	9.60	22.72	72.82

All entries are percentages of rank range or, for the last column, percentage of focal papers. Focal papers are not independent inferential units.

Table S6: Complete factorized tolerance grid.

	Joint source-contrast/selection criterion									Rank criterion		
	$G < .05$ SD			$G < .10$ SD			$G < .15$ SD			Q (%)		
Condition	2.5	5	10	2.5	5	10	2.5	5	10	.5	1	2
ACL												
Random 10%	+	+	−	+	+	−	+	+	−	+	+	+
Source-preferred 10%	+	+	−	+	+	−	+	+	−	+	+	−
Lexical-proxy 10%	+	−	−	+	−	−	+	−	−	+	+	+
Random 25%	+	+	+	+	+	+	+	+	+	+	+	+
Source-preferred 25%	+	+	+	+	+	+	+	+	+	+	+	+
Lexical-proxy 25%	+	+	+	+	+	+	+	+	+	+	+	+
Random 50%	+	+	+	+	+	+	+	+	+	+	+	+
Source-preferred 50%	+	+	+	+	+	+	+	+	+	+	+	+
Lexical-proxy 50%	−	−	−	+	+	+	+	+	+	+	+	+
Biomedical												
Random 25%	+	+	+	+	+	+	+	+	+	+	+	+
Source-preferred 25%	−	−	−	−	−	−	−	−	−	+	+	+
Random 50%	+	+	+	+	+	+	+	+	+	+	+	+
Source-preferred 50%	−	−	−	−	−	−	−	−	−	+	+	+

The nine selection columns give T thresholds in percent, repeated within each displayed G group. + denotes the criterion supported by its original interval; − denotes unsupported. Every one of the 27 joint-with-rank combinations is obtained by combining the appropriate G/T cell with one of the three Q cells. The unabridged 351-row grid is supplied as CSV in Online Resource 2; no thresholds were retuned.

At 10% ACL masking, approximately 83%–88% of mean selection losses lie within the narrow rank neighborhood. This supports a boundary-sensitivity interpretation rather than pervasive rank disorder. At higher rates, losses extend outside that neighborhood. Biomedical outside-neighborhood losses are 6.11%/9.22% of k for random/source masking at 25%, and 18.15%/23.81% at 50%. The continuous movement summaries report the mean across masks of within-mask median movement, 90th-percentile movement and fraction exceeding five percentile points. An elevated Q does not imply substantial movement for every paper, and none of these losses denotes misclassification against independently validated originality.

The frozen sensitivity grid uses aggregate tolerances $\{0.05, 0.10, 0.15\}$, selection tolerances $\{0.025, 0.05, 0.10\}$ and rank tolerances $\{0.005, 0.01, 0.02\}$. It reuses the same simultaneous intervals rather than selecting new preferred thresholds. At a 10% selection tolerance, all three 10%-masking ACL conditions fail the joint rule; at an aggregate tolerance of 0.05 SD, the 50% lexical condition becomes unresolved. Continuous effects and cutoff evidence therefore carry more information than the primary pass/fail labels alone.

S7 The unsuccessful exploratory diagnostic

A pilot ordinary-least-squares diagnostic predicted absolute source-contrast distortion, training on 120 random/source masks and holding out 60 whole lexical-proxy masks. Each local-coverage feature used the fraction of unavailable abstracts among the 20 most similar historical titles to a focal title. The composition model added the absolute difference in missing fractions between sources and between historical years; the local model additionally used the absolute between-focal-source difference in mean local missing fractions and the overall mean local missing fraction. Features were standardized using training data only, with an intercept and least-squares coefficients; baseline scores and hidden abstracts were evaluation targets, not features. Global coverage alone produced held-out MAE 0.02444 SD and RMSE 0.03313. Composition features produced MAE 0.02489 and RMSE 0.03306; adding title-local coverage produced MAE 0.02466 and RMSE 0.03264. The local feature set did not improve MAE over global coverage. The small RMSE difference is insufficient evidence for a useful calibrated diagnostic. There was no feature or hyperparameter tuning on the holdout, no externally validated decision threshold, and no confirmatory or biomedical diagnostic refit. This negative exploratory result is retained to delimit what the project established.

S8 Reproducibility specifications and text rights

The numerical verification compared exact-score calculations, all saved assignment endpoints and family-adjusted intervals with separate implementations. Selection and rank endpoints agreed exactly; small source-contrast differences were within the prespecified floating-point reconciliation tolerance. Reproduction from frozen embeddings regenerated the scoring and numerical readouts. These checks establish computational agreement, while a fresh acquisition and clean-machine encoder rerun were not performed. Full verification scopes and receipts are documented in the reproducibility materials.

The original and revision-stage research, reporting and figure-generation code are archived in version 1.1.0 in Zenodo (Lin, 2026) (`https://doi.org/10.5281/zenodo.23242635`). The additive version 1.1.1 release also supplies the independent scalar-reconstruction script, its runner and tests alongside the 37-file package. Earlier immutable releases retain their original contents. Reporting and saved-membership checks, and local independent scalar-to-endpoint reconstruction, have been verified. Complete source acquisition and encoder reconstruction remain incomplete. Supplementary Sections S8 and S13.6 distinguish these layers.

Online Resource 2 contains original and revision-stage derived numerical summaries, identifier projections and saved cutoff/loss-membership records. Complete masked scalar vectors, ordered identifiers, original masks and paired assignment mappings are publicly available in the 37-file `verification/scalar_scores` package of the version 1.1.1 release. Focal-neighbor records remain private. Source texts, embeddings, model weights and licensed literature are excluded. Grants cover only the explicitly mapped author-controlled files, not third-party inputs. Reporting, saved membership, complete scalar endpoints, saved-embedding scoring and full acquisition/encoding remain distinct routes; downstream reconstruction does not establish an upstream rerun.

Reacquisition and text rights

The original encoder is `allenai/specter`, revision `81cbfb43d4fc2e728d5b4201ce14987db8d0854c`. The original numerical environment used Python 3.12.14, NumPy 2.3.5, PyTorch 2.8.0, transformers 4.57.1 and mpmath 1.3.0. Ordered input identifiers and hashes, seed vectors, the pinned dependency lock, model/tokenizer manifests and locally frozen protocols identify the reported computation. The seed algorithm is specified in Section S3; the separate SPECTER2 settings are in Section S12.

Reproduction requires the ordered inputs, declared eligibility rules, pinned model and preprocessing settings,

and recorded assignments. Reacquisition may differ because metadata, retraction flags, API behavior or access conditions change; the retained snapshot defines the reported results. Source texts and licensed literature copies require a rights-aware access plan separate from release of code and numerical summaries.

Biomedical records preserve authors, source links and item rights statements. Of 4,862 retained records, 4,652 statements identify Creative Commons Attribution and 210 identify CC0/public-domain dedication. No returned item statement specifies a numbered license version, which is therefore not inferred. Absence of a PubMed copyright field is not permission. Raw abstract or embedding redistribution requires a separate rights and release decision. The item-level rights statements and acquisition receipts are retained with the reproducibility materials.

S9 Prior-art boundary

The general difference between aggregate and individual stability predates this study. Reference-pool sensitivity, rank perturbation, title-versus-longer-input comparisons and uncertainty propagation also have direct precedents. The scoped contribution combines controlled historical title substitution with use-specific and representation-specific evaluation: fixed focal papers, exact missing counts, paired opportunity controls, separate aggregate/rank/selection outcomes, cutoff qualification in two selected collections, and a limited related-encoder sensitivity. It is not a new novelty indicator or a demonstration that the baseline measures true scientific originality.

The literature review was updated through 6 October 2026. Wang, Ding and Pan's final main text and Appendices A–G were obtained through ordinary publisher access and inspected. Their study uses abstract-only representations, an earlier-paper novelty defined as one minus mean cosine similarity to ten earlier papers, approximate-neighbor fidelity checks, group comparisons and journal-network perturbations. It is therefore close prior art for several components; the inspected text does not report the same historical title-substitution experiment with paired controls and the joint outcomes here.

SemNovel's final full methods and any supplement remain unverified. Its published abstract and preview support aggregate and unusual-paper validation claims, but earlier author-released code cannot establish the scope of the final article. This remaining access limitation qualifies the claim of a narrow empirical extension; absence from inspected code is not evidence that an unread article omits an experiment. The review is bounded, not proof of exhaustive originality. Related input-ablation, reference-composition and ranking-robustness studies are discussed in the main Related Work section with their inspected versions recorded in the source matrix.

For interpretation, the principal limitations are selected complete current frames, finite history windows, one primary pretrained encoder with a limited related-model sensitivity, synthetic interventions, operational tolerances and conditional mask uncertainty. Small source-contrast changes may reflect cancellation, and fixed-size selection may magnify near-cutoff changes. The source-effect difference is informative about limits of transport across these collections, but its cause is unidentified. The current coverage audit neither removes these limitations nor resolves SemNovel's methods uncertainty.

S10 Post-study OpenAlex coverage audit

A current exact-DOI audit on 6 October 2026 found candidates for all 7,190 historical DOIs. Of these, 3,787 ACL/EMNLP and 3,389 biomedical DOIs had unique matches with a valid, nonempty abstract index. Twelve ACL/EMNLP and two biomedical DOIs had ambiguous duplicate matches. No queried DOI was unmatched. Among the 7,176 uniquely matched DOIs, none lacked the requested abstract field, returned an explicit null, or had a malformed index. Known-null counts were zero in all twelve source-year cells. Duplicate candidates also had nonempty indexes, although some contained only one or two terms, illustrating why nonempty availability is not semantic integrity.

The prespecified requirement of 100% unique matches failed for 14 DOIs. Moreover, no observed abstract omissions were available to define an informative empirical masking pattern. Consequently no score-mask experiment followed this audit. It supplies a current coverage check, not real-missingness validation, evidence of historical availability, or empirical calibration of 10%, 25% and 50% synthetic rates. A nonempty inverted index was not validated for completeness, correspondence to the original abstract or usefulness for the encoder.

Requests selected DOI, work identifier, publication year and abstract-index state, using fixed batches of 100

exact normalized DOIs with capacity for 200 returned works. Complete responses, request identities and count reconciliation were retained. The retrieval date was 6 October 2026; exact timestamps, acquisition-completeness repair and verification receipts are documented in the reproducibility materials.

Table S7: Dated OpenAlex availability among the fixed historical records.

Source	Year	Frame	Unique present	Null	Multiple
ACL					
ALL	ALL	3799	3787	0	12
ACL	2018	381	378	0	3
ACL	2019	660	654	0	6
ACL	2020	778	776	0	2
EMNLP	2018	548	548	0	0
EMNLP	2019	681	681	0	0
EMNLP	2020	751	750	0	1
Biomedical					
ALL	ALL	3391	3389	0	2
PLOSCB	2018	528	527	0	1
PLOSCB	2019	620	620	0	0
PLOSCB	2020	706	706	0	0
PLOSGEN	2018	536	536	0	0
PLOSGEN	2019	496	496	0	0
PLOSGEN	2020	505	504	0	1

No unmatched, field-omitted or malformed unique indexes occurred. Multiple-DOI states remain separate from unavailable abstracts. A valid nonempty index denotes exposed field structure, not correctness of the abstract. Null-rate denominators are unique known field states; counts condition on the fixed complete-text frames.

S11 Post-confirmation studentized bootstrap sensitivity

The original Student inference remains unchanged. A separately frozen post hoc sensitivity resampled all 200 whole masks within each condition, carrying G, T, Q together and retaining each structured/control pair. The mean paired effect was formed before resampling; focal papers were not resampled as independent observations. There was no sign-flip randomization, because structured and control arms are not exchangeable. Moderate observed skewness and discrete selection losses motivated the sensitivity: maximum absolute skewness was 1.065 in ACL and 1.178 in the biomedical vectors.

The fixed plan used 200,000 resamples per condition and root seed 20261006. The first 16 bytes of SHA-256 for the compact JSON array [`major-revision-studentized-bootstrap-v1`, root seed, study, rate, mechanism] were interpreted as a big-endian integer seed for PCG64. All 200 whole-mask rows were sampled with replacement in each resample. For each endpoint, $t^* = (\bar{x}^* - \bar{x})/(s^*/\sqrt{200})$ and the interval was $[\bar{x} - q_{1-a}\, s/\sqrt{200}, \bar{x} - q_a\, s/\sqrt{200}]$, with $a = .05/(2J)$ and $J = 45$ or 18. Quantiles used linear interpolation; no interval clipping, jitter or adaptive extension was applied. The sensitivity retained the separate original inferential scopes. Each extreme tail has approximately 111 ACL or 278 biomedical resamples, so finite-resampling uncertainty matters. Approximate binomial-order-statistic bands describe Monte Carlo uncertainty in endpoints; they are pointwise computational diagnostics, not additional scientific confidence intervals or a joint guarantee.

All 63 classifications/directions calculated using the estimated bootstrap quantiles agreed with the original Student classifications, and all thirteen joint G/T classifications agreed. ACL source 10% T had bootstrap interval [5.0293%,5.7834%], compared with the original lower bound 5.0112%. The new lower endpoint's approximate Monte Carlo band was [5.0224%,5.0323%]. This case still remains practically close to the operational cutoff.

The small ACL lexical 10% paired Q direction was less secure: its lower endpoint's approximate Monte Carlo band spanned −0.000261 to +0.002818 percentage points. The estimated interval had a positive lower bound, but the direction should not be presented as robust to finite-bootstrap tail error. There were no joint-rule changes. The sensitivity does not establish exact finite-sample coverage, population validity or a new three-study family.

All 63 original and sensitivity intervals, endpoint Monte Carlo bands and classifications are supplied in Online Resource 2 as `interval_sensitivity.csv`; `joint_sensitivity.csv` supplies the thirteen joint decisions. The

frozen plan, input hashes and complete resampling seed identities are documented in the reproducibility materials.

S12 Post-confirmation related-encoder sensitivity

This post hoc check used the pinned SPECTER2 base with its activated proximity adapter (Singh et al., 2023; full reference in the main article). It reused every original 25% random assignment in both collections and every original biomedical 25% source-preferred assignment and matched control (200 per condition). Baselines, population SDs, source contrasts, ranks and selected sets were recomputed within SPECTER2. No primary masks, thresholds or results were changed. The 12 reported means form a new, separate approximate 95% Bonferroni–Student family, with df=199 and critical value 2.8987694543; there is no pooled or cross-encoder significance test.

Both random conditions retain low distortion under the operating 0.10-SD tolerance alongside selection replacement exceeding 5% and rank displacement exceeding 1%. The biomedical source-preferred condition exceeds the aggregate threshold, and all three paired source-minus-control means are positive. Magnitudes differ across encoders. More fundamentally, the biomedical baseline source contrast changes from $+.2613s_0$ to $-.0076s_0$, so transport of the threshold pattern does not establish stability or validity of the journal comparison. The aggregate tolerance exceeds the absolute SPECTER2 baseline contrast in both collections. This related encoder, three-condition check does not assess ACL structured arms, other missingness rates, other measures or encoder-independent novelty validity.

Table S8: Baseline quantities calculated within each encoder. The cutoff is the midpoint between the kth and $(k+1)$th descending baseline scores.

Collection	Encoder	Population SD	Source contrast/s_0	Cutoff	Gap/s_0
ACL/EMNLP	SPECTER	0.027866	-0.130701	0.137822	0.000896
ACL/EMNLP	SPECTER2	0.012892	-0.096829	0.066434	0.000416
Biomedical	SPECTER	0.032666	+0.261287	0.154532	0.004025
Biomedical	SPECTER2	0.013068	-0.007625	0.084672	0.002823

Source contrast is mean ACL minus mean EMNLP, or mean PLOS Computational Biology minus mean PLOS Genetics, before masking. Its SD uses the full focal baseline cohort. Encoder-specific scales and memberships are never reused across models. The baseline boundary contains 16 selected papers in the fixed rank band in each SPECTER2 collection (band sizes 31 and 30).

Table S9: SPECTER2 means and separate 12-family intervals at 25% historical masking. T and Q are percentages; their paired differences are percentage points. G is in the collection's own baseline SD units.

Collection	Assignment	Endpoint	Mean	Interval	Expected lost
ACL/EMNLP	Random	G	0.0146	[0.0125, 0.0166]	—
ACL/EMNLP	Random	T	12.1058	[11.6824, 12.5292]	18.885
ACL/EMNLP	Random	Q	4.6907	[4.6209, 4.7604]	—
Biomedical	Random	G	0.0170	[0.0141, 0.0199]	—
Biomedical	Random	T	17.2264	[16.7227, 17.7300]	25.495
Biomedical	Random	Q	5.3878	[5.3111, 5.4645]	—
Biomedical	Source-preferred	G	0.1220	[0.1183, 0.1257]	—
Biomedical	Source-preferred	T	20.4122	[19.8617, 20.9626]	30.210
Biomedical	Source-preferred	Q	6.2278	[6.1389, 6.3168]	—
Biomedical	Source minus control	ΔG	0.1053	[0.1008, 0.1099]	—
Biomedical	Source minus control	ΔT	3.1588	[2.3767, 3.9409]	—
Biomedical	Source minus control	ΔQ	0.8618	[0.7452, 0.9784]	—

G is absolute source-contrast distortion before averaging; paired ΔG compares those absolute distortions and is not a signed journal contrast. Paired endpoints are structured minus their original year-matched random controls. Expected lost selections are mean $T \times k$ ($k = 156$ or 148); they describe mask expectations rather than fractional individual papers. No combined 95% guarantee with the primary 45/18 families is claimed.

SPECTER2 full inputs were truncated for 43 biomedical papers (26 historical; 17 focal), compared with 67 under the original encoder. No biomedical title inputs or ACL inputs were truncated. The effects of encoder

choice and truncation were not separated. Numerical verification reconciled all 800 saved base/control score vectors, 200 paired rows and twelve mean intervals. Execution and verification receipts are documented in the reproducibility materials.

The base identifier is `allenai/specter2_base`, revision `3447645e1def9117997203454fa4495937bfbd83`; the proximity adapter is `allenai/specter2`, revision `2081559630a80fc5851d8f798a05ba81e9468089`. CLS pooling, L2 normalization, float32 output and the 512-token cap were retained. Inference used evaluation mode with the proximity adapter active, four CPU threads and batches of eight. The software environment used transformers 4.57.6 and adapters 1.3.0, with Python 3.12.14, PyTorch 2.8.0 and NumPy 2.3.5. The original SPECTER environment was unchanged.

Online Resource 2 includes `primary_and_paired.csv`, `readout.json` and encoder-specific comparison tables. Recalculation also requires the ordered source inputs and each encoder's full/title arrays; numerical tables alone are not a complete input bundle. Model file hashes, executed source versions, separate amendments and reconstruction instructions are supplied in the reproducibility materials.

S13 Additional descriptive analyses and computational verification

These post hoc analyses were specified after inspection of the original experimental outcomes and critical review of the manuscript. They preserve the original cohorts, masks, seeds, encoder outputs, endpoints, operating tolerances and inference families. They describe the fixed stored outcomes and do not constitute a new confirmatory experiment. Full-precision summaries are separate in Online Resource 2; the scalar package is supplied separately in version 1.1.1; focal-neighbor records remain private.

S13.1 Signed contrasts, level drift, and source-specific shifts

These descriptive summaries use frozen results and remain outside the original simultaneous inference families. Let b_i and z_{im} denote the complete-history and masked scores, let s_0 be the original whole-cohort population standard deviation, and let $w = n_A/n$. Define

$$C_0 = \{\bar{b}_A - \bar{b}_B\}/s_0, \qquad C_m = \{\bar{z}_{A,m} - \bar{z}_{B,m}\}/s_0, \tag{1}$$

$$D_m = C_m - C_0, \qquad M_m = \overline{(z_m - b)}/s_0, \tag{2}$$

$$A_m = \overline{(z_m - b)}_A/s_0, \qquad B_m = \overline{(z_m - b)}_B/s_0. \tag{3}$$

Thus $D_m = A_m - B_m$, $M_m = wA_m + (1-w)B_m$, $A_m = M_m + (1-w)D_m$, and $B_m = M_m - wD_m$. Here source A/B means ACL/EMNLP or PLOS Computational Biology/PLOS Genetics, and the weights are 710/1557 and 934/1471, respectively. Equal weighting of sources would not recover the whole-cohort mean. The primary endpoint remains $G_m = |D_m|$: in general $\overline{G} \neq |\overline{D}|$, and a paired difference in G is not a signed contrast change.

Tables S10 and S11 are generated from full-precision mixed-policy output rows, with deterministic restored and all-title rows excluded. Mean source shifts are first reconstructed algebraically from saved signed changes and drift; direct calculation from the original complete masked scalar vectors independently agrees across all 4,200 original arms (maximum absolute difference below 5×10^{-16} baseline SD). Each condition contains the original 200 assignments. The tables report descriptive means without new simultaneous intervals; assignment variability for the original endpoints remains in Tables S3–S4 and the original numerical files.

Under random 25% substitution, the ACL/EMNLP source shifts are +0.1180 and +0.1021 baseline SD; the biomedical shifts are +0.1076 and +0.1185. Common upward movement therefore partly cancels in the source contrast. Mean whole-cohort drift is +0.1094 and +0.1116 SD, whereas mean absolute contrast distortion is 0.0175 and 0.0167 SD. Biomedical source-preferred substitution gives source shifts of +0.1842 versus +0.0356 SD at 25%, and +0.4735 versus +0.0965 SD at 50%. Its mean contrast consequently rises from the complete-history +0.2613 SD to +0.4099 and +0.6383 SD, increases of 56.9% and 144.3% relative to this positive observed baseline. This is inflation of the calculated source contrast, not a masking-induced reversal or a claim about true disciplinary originality. The algebra by itself does not identify the semantic mechanism.

Table S10: Additional descriptive means for all 13 original base conditions, in the original baseline SD. Source and lexical denote the original preferred-substitution mechanisms; lexical is the title-substring proxy. C is the signed contrast; D its signed change; G absolute distortion; M whole-cohort drift; A, B source shifts.

Collection	Condition	$\overline{C}$	$\overline{D}$	$\overline{G}$	$\overline{M}$	$\overline{A}$	$\overline{B}$
ACL	Random 10%	-0.1249	+0.0058	+0.0088	+0.0407	+0.0438	+0.0381
ACL	Random 25%	-0.1148	+0.0159	+0.0175	+0.1094	+0.1180	+0.1021
ACL	Random 50%	-0.0963	+0.0344	+0.0352	+0.2672	+0.2859	+0.2515
ACL	Source 10%	-0.1285	+0.0022	+0.0069	+0.0375	+0.0387	+0.0365
ACL	Source 25%	-0.1215	+0.0092	+0.0129	+0.1039	+0.1090	+0.0997
ACL	Source 50%	-0.1034	+0.0273	+0.0277	+0.2532	+0.2681	+0.2408
ACL	Lexical 10%	-0.1199	+0.0108	+0.0114	+0.0411	+0.0469	+0.0361
ACL	Lexical 25%	-0.1004	+0.0303	+0.0305	+0.1259	+0.1424	+0.1121
ACL	Lexical 50%	-0.0837	+0.0470	+0.0471	+0.3011	+0.3267	+0.2797
Biomedical	Random 25%	+0.2504	-0.0109	+0.0167	+0.1116	+0.1076	+0.1185
Biomedical	Random 50%	+0.2395	-0.0218	+0.0267	+0.2645	+0.2566	+0.2784
Biomedical	Source 25%	+0.4099	+0.1486	+0.1486	+0.1300	+0.1842	+0.0356
Biomedical	Source 50%	+0.6383	+0.3770	+0.3770	+0.3359	+0.4735	+0.0965

Table S11: The same descriptive quantities for the eight matched-control conditions, kept separate from the base assignments. Each control is paired with its corresponding structured assignment; it is not an independent scientific replication.

Collection	Condition	$\overline{C}$	$\overline{D}$	$\overline{G}$	$\overline{M}$	$\overline{A}$	$\overline{B}$
ACL	Source control 10%	-0.1256	+0.0051	+0.0083	+0.0413	+0.0441	+0.0390
ACL	Source control 25%	-0.1163	+0.0144	+0.0170	+0.1136	+0.1214	+0.1071
ACL	Source control 50%	-0.0973	+0.0334	+0.0343	+0.2726	+0.2907	+0.2573
ACL	Lexical control 10%	-0.1246	+0.0061	+0.0091	+0.0400	+0.0434	+0.0372
ACL	Lexical control 25%	-0.1179	+0.0128	+0.0155	+0.1098	+0.1167	+0.1040
ACL	Lexical control 50%	-0.0989	+0.0318	+0.0323	+0.2663	+0.2836	+0.2519
Biomedical	Source control 25%	+0.2512	-0.0101	+0.0171	+0.1120	+0.1084	+0.1185
Biomedical	Source control 50%	+0.2391	-0.0221	+0.0285	+0.2649	+0.2568	+0.2790

S13.2 Capacity, nested sets, and conditional selection loss

Let S_0 be the original selected set, B the original tie-expanded rank band, and $\mathcal{L}_m = S_0 \setminus S_m$ the lost selections. Write $L_m = |\mathcal{L}_m|$, $I_m = |\mathcal{L}_m \cap B|$, $O_m = |\mathcal{L}_m \setminus B|$, and $b_{\text{band}} = |S_0 \cap B|$. Then $L_m = I_m + O_m$, $0 \le I_m \le b_{\text{band}}$, and

$$\overline{O} \;\ge\; \overline{(L - b_{\text{band}})_+} \;\ge\; (\overline{L} - b_{\text{band}})_+, \qquad (x)_+ = \max(0, x). \tag{4}$$

The original band contains 16 baseline selections in each collection. All per-assignment partitions, bounds and original count projections were checked for the 4,200 original arms. Table S12 reports both lower bounds for base conditions; matched controls remain available in the numerical resource, with individual assignments available in the public scalar package. For ACL random 50%, mean loss is 42.0 papers, mean outside-band loss 30.4, and both bounds 26.0. For biomedical source-preferred 50%, the corresponding values are 47.4, 35.2, and 31.4. These inequalities constrain interpretation of outside-band counts; they are not a random-loss model, significance test, correction factor or measure of scientific harm. The separate whole-cohort rank-movement results retain their own interpretation.

The strict positive-margin set at > 0.10 SD is nested within the outside-band selected set in ACL (132 within 140 papers). In biomedical data the outside-band set is nested within the strict margin set (132 within 133 papers). Their intersections contain 132 papers in both collections; the respective set differences contain eight ACL papers and one biomedical paper. These related summaries must not be added or treated as independent corroboration.

Table S12: Original base-condition mean losses and capacity lower bounds, in expected paper counts across 200 assignments. $(x)_+ = \max(0, x)$; capacity is 16 baseline selections in both collections. Displayed zero values may reflect rounding of small nonzero quantities; full precision is retained in the numerical resource.

Collection	Condition	$\overline{L}$	$\overline{O}$	$\overline{(L-16)_+}$	$(\overline{L}-16)_+$
ACL	Random 10%	9.7	1.2	0.0	0.0
ACL	Random 25%	22.0	9.1	6.0	6.0
ACL	Random 50%	42.0	30.4	26.0	26.0
ACL	Source 10%	8.4	1.0	0.0	0.0
ACL	Source 25%	19.6	8.1	3.8	3.6
ACL	Source 50%	37.8	28.1	21.8	21.8
ACL	Lexical 10%	7.2	1.2	0.0	0.0
ACL	Lexical 25%	20.8	7.7	4.9	4.8
ACL	Lexical 50%	42.5	30.5	26.5	26.5
Biomedical	Random 25%	22.1	9.0	6.1	6.1
Biomedical	Random 50%	37.8	26.9	21.8	21.8
Biomedical	Source 25%	26.5	13.6	10.5	10.5
Biomedical	Source 50%	47.4	35.2	31.4	31.4

Conditional loss uses the unchanged signed-margin edges

$$(-\infty, -.10, -.05, -.025, -.01, 0, .01, .025, .05, .10, +\infty),$$

with left-closed/right-open bins. For a fixed bin A_b, its denominator is $N_b = |S_0 \cap A_b|$ and its mean loss probability is $\sum_m |\mathcal{L}_m \cap A_b|/(200N_b)$. Exact zero belongs to $[0, .01)$; empty bins are undefined. The last displayed bin includes .10, whereas the original far-margin diagnostic uses strict $> .10$; these rules are retained rather than conflated. Table S13 supplies all positive-bin denominators. Negative bins contain no baseline-selected papers in these cohorts and are retained as undefined rows in the numerical outputs.

The probabilities computed from original lost-ID records agree with averages of the original per-paper selection probabilities over the same baseline-selected IDs and with the saved bin-retention summaries. In every condition, $\sum_b N_b \overline{p}_{\text{loss},b}$ recovers total mean selection losses. Main conditional-loss curves display the shared random/source-preferred 25% and 50% conditions in both collections; the remaining ACL conditions are shown in the supplement. All 13 base conditions and eight controls remain in `conditional_loss_by_margin.csv`. These are descriptive conditional probabilities: no original total-loss interval is transferred to a bin, and no paper–mask independence or binomial interval is assumed.

Table S13: Fixed baseline-selected denominators in the positive signed-margin bins. The rows sum to 156 ACL and 148 biomedical selections.

Collection	$[0, .01)$	$[.01, .025)$	$[.025, .05)$	$[.05, .10)$	$[.10, \infty)$
ACL	2	4	5	13	132
Biomedical	3	1	2	9	133

S13.3 Complete-history agreement between encoder pipelines

This post hoc descriptive comparison uses the four already saved complete-history scalar baselines; it performs no new encoding. Canonical focal-ID alignment is exact within collection (1,557 ACL/EMNLP and 1,471 biomedical papers), with no duplicate IDs or dropped records. Source labels agree with the original frozen record mappings. All 6,056 projected baseline scalar values exactly match their original saved vectors. Each encoder uses its own population SD, exact-score average ranks and top $\lceil .10n \rceil$ set with the original lexical-ID tie rule.

Define ρ_{cross} as the Pearson correlation of the two average-rank vectors, $Q_{\text{cross}} = \text{mean}_i\, |r_{i,\text{SPECTER}} - r_{i,\text{SPECTER2}}|/(n-1)$, and overlap as $|S_{\text{SPECTER}} \cap S_{\text{SPECTER2}}|/k$. No no-ties shortcut is used. Table S14 shows correlations of 0.684 and 0.589, normalized movement of 0.174 and 0.202, and shared selections of 81/156 (51.9%) and 62/148 (41.9%). Thus the related pipelines differ materially in complete-history ranking and individual selection, alongside the previously reported selected-condition perturbation responses.

Selected source composition changes less than individual membership (Table S15): ACL counts are 64 versus 60 under SPECTER/SPECTER2, and PLOS Computational Biology counts are 106 versus 97. These proportions use k as denominator; within-source selection rates use the full source cohort and are recorded separately in `cross_encoder_source_composition.csv`. The biomedical complete-history contrast is positive under SPECTER (+0.2613 own-baseline SD) and approximately zero, slightly negative under SPECTER2 (−0.0076 own-baseline SD). The latter is not a large opposite substantive effect. Because the pipelines also differ in truncation, these differences cannot be attributed solely to encoder identity. Neither within-pipeline perturbation sensitivity nor between-pipeline agreement validates true scientific originality.

Table S14: Post hoc complete-history agreement between SPECTER and SPECTER2. Retained/replaced refer to cross-encoder overlap within each fixed top-decile set; they are not additional masking outcomes.

Collection	ρ_{cross}	Q_{cross}	Overlap / k	Retained	Replaced
ACL	0.684	0.174	81/156	51.9%	48.1%
Biomedical	0.589	0.202	62/148	41.9%	58.1%

Table S15: Source composition of each encoder's complete-history top-decile set. The denominator is the collection-specific k for all percentages.

Collection	Encoder	Source	Selected	Share of top k
ACL	SPECTER	ACL	64	41.0%
ACL	SPECTER	EMNLP	92	59.0%
ACL	SPECTER2	ACL	60	38.5%
ACL	SPECTER2	EMNLP	96	61.5%
Biomedical	SPECTER	PLOS Comput. Biol.	106	71.6%
Biomedical	SPECTER	PLOS Genetics	42	28.4%
Biomedical	SPECTER2	PLOS Comput. Biol.	97	65.5%
Biomedical	SPECTER2	PLOS Genetics	51	34.5%

S13.4 Realized intensity of the original interventions

These post hoc descriptive summaries use the saved original missing-history IDs for every base and matched-control assignment; no new masks were drawn. For historical stratum s, let N_s denote its fixed size and c_{ms} the number substituted in assignment m. The within-stratum unavailable fraction is $f_{ms} = c_{ms}/N_s$, whereas

the stratum's share of the masked set is $c_{ms}/\sum_s c_{ms}$. These have different denominators. The original 8:1 preference applies to exponential-clock sampling rates, not to marginal inclusion probabilities or odds. Tables S16 and S17 report the realized fractions. The numerical resource reports historical denominators, count summaries, composition shares, means, sample assignment SDs, and 5th/95th percentiles using linear interpolation (NumPy's `method='linear'`). Each condition has 200 assignments; the source and lexical matched controls remain separate.

At nominal 25% loss, source-preferred substitution masks on average 44.4% of ACL and 7.2% of EMNLP histories, compared with 40.5% of PLOS Computational Biology and 6.3% of PLOS Genetics histories. For example, ACL histories constitute 85.1% of the ACL collection's masked set under this condition, which is distinct from their 44.4% within-source substitution fraction. The ACL lexical preference affects the original 289 titles matching the case-insensitive substring `translat`: their mean substituted fraction is 83.8% at nominal 25% loss and 99.2% at nominal 50% loss. This describes the synthetic intervention, not estimated real-world abstract missingness.

Exact total mask counts and every original matching cell reconcile across all 4,200 scored arms (2,600 base assignments and 1,600 controls). Source controls match by historical year; lexical controls match by historical year and source. Lexical membership comes from the frozen record fields and agrees with the original title substring rule; no contemporary titles or inferred topic labels were substituted.

Table S16: Realized historical substitution within source strata (percent). All entries use the original 200 assignments. Parentheses give assignment SD. Source A/B denotes ACL/EMNLP or PLOS Computational Biology/PLOS Genetics; N_A, N_B are historical source sizes. Percentiles and masked-set composition are in the numerical resource.

Collection	Condition	Loss	Source A	Source B	N_A	N_B
ACL	Random	10	10.0 (0.5)	10.0 (0.5)	1819	1980
ACL	Random	25	25.0 (0.7)	25.0 (0.6)	1819	1980
ACL	Random	50	50.1 (0.9)	50.0 (0.8)	1819	1980
ACL	Source	10	18.2 (0.4)	2.5 (0.3)	1819	1980
ACL	Source	25	44.4 (0.6)	7.2 (0.5)	1819	1980
ACL	Source	50	82.9 (0.7)	19.8 (0.6)	1819	1980
ACL	Source control	10	10.0 (0.4)	10.0 (0.4)	1819	1980
ACL	Source control	25	25.1 (0.8)	24.9 (0.7)	1819	1980
ACL	Source control	50	50.2 (0.9)	49.8 (0.8)	1819	1980
ACL	Lexical	10	10.2 (0.5)	9.8 (0.4)	1819	1980
ACL	Lexical	25	25.4 (0.7)	24.7 (0.6)	1819	1980
ACL	Lexical	50	50.3 (0.8)	49.7 (0.8)	1819	1980
ACL	Lexical control	10	10.2 (0.5)	9.8 (0.4)	1819	1980
ACL	Lexical control	25	25.4 (0.7)	24.7 (0.6)	1819	1980
ACL	Lexical control	50	50.3 (0.8)	49.7 (0.8)	1819	1980
Biomedical	Random	25	25.0 (0.7)	25.0 (0.8)	1854	1537
Biomedical	Random	50	50.0 (0.8)	50.0 (1.0)	1854	1537
Biomedical	Source	25	40.5 (0.5)	6.3 (0.6)	1854	1537
Biomedical	Source	50	77.4 (0.7)	17.0 (0.8)	1854	1537
Biomedical	Source control	25	25.1 (0.7)	24.9 (0.9)	1854	1537
Biomedical	Source control	50	50.1 (0.9)	50.0 (1.0)	1854	1537

Displayed zero values may reflect rounding of small nonzero quantities; full precision is retained in the numerical resource.

Table S17: Realized substitution in the ACL lexical-proxy strata (percent). Entries are means (assignment SDs) over 200 masks. The proxy stratum comprises 289 original titles matching the case-insensitive substring `translat`; the nonproxy stratum comprises 3,510 histories.

Condition	Loss	Proxy	Nonproxy
Lexical	10	45.0 (2.5)	7.1 (0.2)
Lexical	25	83.8 (2.3)	20.2 (0.2)
Lexical	50	99.2 (0.5)	46.0 (0.0)
Lexical control	10	10.2 (1.8)	10.0 (0.1)
Lexical control	25	25.2 (2.4)	25.0 (0.2)
Lexical control	50	50.3 (2.9)	50.0 (0.2)

Displayed zero values may reflect rounding of small nonzero quantities; full precision is retained in the numerical resource.

S13.5 Post hoc exact-neighbor accounting

This bounded analysis uses the existing original SPECTER full/full and full/title exact similarity matrices, with the original focal/history ordering, preprocessing, normalization, temporal eligibility and float32 score arithmetic. Its fixed scope is source-preferred 25% and 50% substitution and year-matched controls in both collections, retaining all 200 pairs per rate and collection. The first eligible assignment in each frozen seed-manifest order was benchmarked before the complete diagnostic. No encoding, new model, new corpus or new assignments were introduced.

For each focal paper and assignment, the diagnostic records the complete-history winning historical identity/source, whether that winner was masked, the perturbed winning identity/source and full/title representation, signed score change, original top-decile membership and subsequent loss. Historical winner identities were not specified by the original maximum-only scorer. For this new diagnostic only, lexical historical IDs resolve exact maxima; tie flags and the complete-history best-to-second-best similarity gap are retained. No exact maximum ties occurred in the analyzed baseline or masked rows. Retaining the same identity while substituting its title differs from changing the identity; a previously nonwinning title can also become the maximum-similarity candidate.

All summaries first use assignment-level fractions with fixed whole-cohort or focal-source denominators. Selected losses use the fixed number of baseline-selected papers in that focal source. Exposure, identity switching and winning representation jointly define eight transition cells. A cell's score-shift contribution is the sum of its standardized changes divided by the fixed focal-source count, so contributions add to that source's mean shift. Conditional cell means instead use that cell's own count; empty cells remain undefined and the number of defined assignments is reported. No focal-mask independence or causal mediation interpretation is assumed.

Table S18: Post hoc exact-neighbor summaries for source-preferred substitution. Values are assignment means over 200 masks; exposure, identity switch and title-winner columns use the fixed focal-source denominator. Matched controls and transition contributions are in the numerical resource.

Collection/source	Masking (%)	Exposed	Switch	Title	Shift	Selected loss (%)
ACL/ACL	25	23.0	22.9	2.1	+0.109	11.8
ACL/EMNLP	25	24.0	24.1	3.3	+0.100	13.2
ACL/ACL	50	46.8	45.7	6.5	+0.268	23.7
ACL/EMNLP	50	48.2	47.1	7.9	+0.241	24.6
Biomedical/PLOSCB	25	38.2	34.9	10.5	+0.184	13.9
Biomedical/PLOSGEN	25	8.7	7.9	1.6	+0.036	28.1
Biomedical/PLOSCB	50	73.4	61.4	33.9	+0.474	23.6
Biomedical/PLOSGEN	50	21.2	18.9	4.4	+0.096	53.3

Exposure, switching, title winner and selected loss are percentages; shift is in baseline population SD units. Selected loss uses the fixed baseline-selected count within each source.

Under biomedical source-preferred 25% substitution, the baseline winner is masked for an average 38.2% of PLOS Computational Biology focal papers and 8.7% of PLOS Genetics focal papers; the matched-control exposures are 25.0% and 25.2%. At 50%, source-preferred exposure rises to 73.4% and 21.2%. Corresponding source

shifts are +0.184 and +0.036 baseline SD at 25%, and +0.474 and +0.096 at 50%. In the ACL collection at 25%, exposure is more similar between focal sources (23.0% and 24.0%) and below the matched controls (25.3% and 25.4%); source shifts are +0.109 and +0.100 SD.

For example, the PLOS Computational Biology shift under source-preferred 25% substitution decomposes into +0.187989 SD from focal papers whose baseline winner is masked and -0.003745 SD from those whose winner remains full, summing to +0.184244 SD. These are fixed-denominator contributions, not conditional subgroup means. The latter contribution can be negative because a newly available title representation can exceed the retained baseline maximum. Conversely, retaining the baseline winning full representation makes an increase in novelty impossible; every analyzed row satisfies this bound.

The cache-derived baseline and masked scores reproduce every saved original scalar exactly in the specified scope. All transition contributions reconcile with the independently reconstructed source shifts in Section S13.1. These findings describe where changes occur in the nearest-neighbor operation and help characterize the contrasting collection responses. They do not establish why the corpora differ semantically, isolate encoder truncation, or identify a causal role of source labels. Assignment-distribution and transition summaries are included in the numerical resource; focal-level neighbor/scalar-change records remain private by the author's decision.

S13.6 Reconstruction from complete saved scalar scores

The public computational-verification package exports the existing mixed-policy scalar records losslessly, without rescoring or encoding: 3,000 SPECTER ACL arms, 1,200 SPECTER biomedical arms, 200 SPECTER2 ACL arms and 600 SPECTER2 biomedical arms. The total is 5,000 scored arms and 7,630,200 masked scalar values, plus 6,056 baseline values. Repeated all-title/restoration policies and the SPECTER2 paired endpoint-only records are not additional scored arms. Four `scores.npz` files contain non-object numerical arrays, loaded with `allow_pickle=False`; float64 storage preserves every parsed saved JSON number exactly. Ordered focal/history identifiers, source labels, mask membership, seeds, original arm identifiers and paired assignment mappings accompany the arrays.

The independently implemented `reconstruct_endpoints_from_scalar_scores.py` uses these scalars rather than saved endpoint columns as its calculation inputs and imports no original endpoint functions. During the prior private revision, a clean extraction reconstructed all 63 original family means and 12 separate SPECTER2 means, their mask/pair SDs, MCSEs and original simultaneous intervals. Selected/lost identifiers, cutoff subsets and pairing identities matched exactly. The historical reconciliation bounds were retained: 2×10^{-5} for the float32-scoring comparison of G and 2×10^{-10} for T, Q; family readout comparisons used 2×10^{-10}. The largest family mean and interval-bound deviations were below 4×10^{-18}, and the largest discrepancy from float32 scoring aggregation was 1.16×10^{-6} for G. Fifteen synthetic fixtures passed. A file-access and network guard permitted only the extraction and installed Python dependencies during the endpoint run. Direct source shifts also agreed with the separate algebraic reconstruction across all 4,200 original arms.

The scalar-to-endpoint route was freshly repeated in a separate private copy during final release preparation: all 5,000 arms, 75 summary rows and 116,898 assertions passed, together with the 15 fixtures. No new scalar scoring or encoding was performed. Its 37-file scalar package is now public in version 1.1.1; the focal-neighbor records remain private. The scalar package is separate from Online Resource 2 and the writing ZIP. Endpoint reconstruction does not reproduce scalar scoring from embeddings or fresh acquisition and encoding. The version 1.0.0 archive contains neither these records nor the revision analysis implementation.

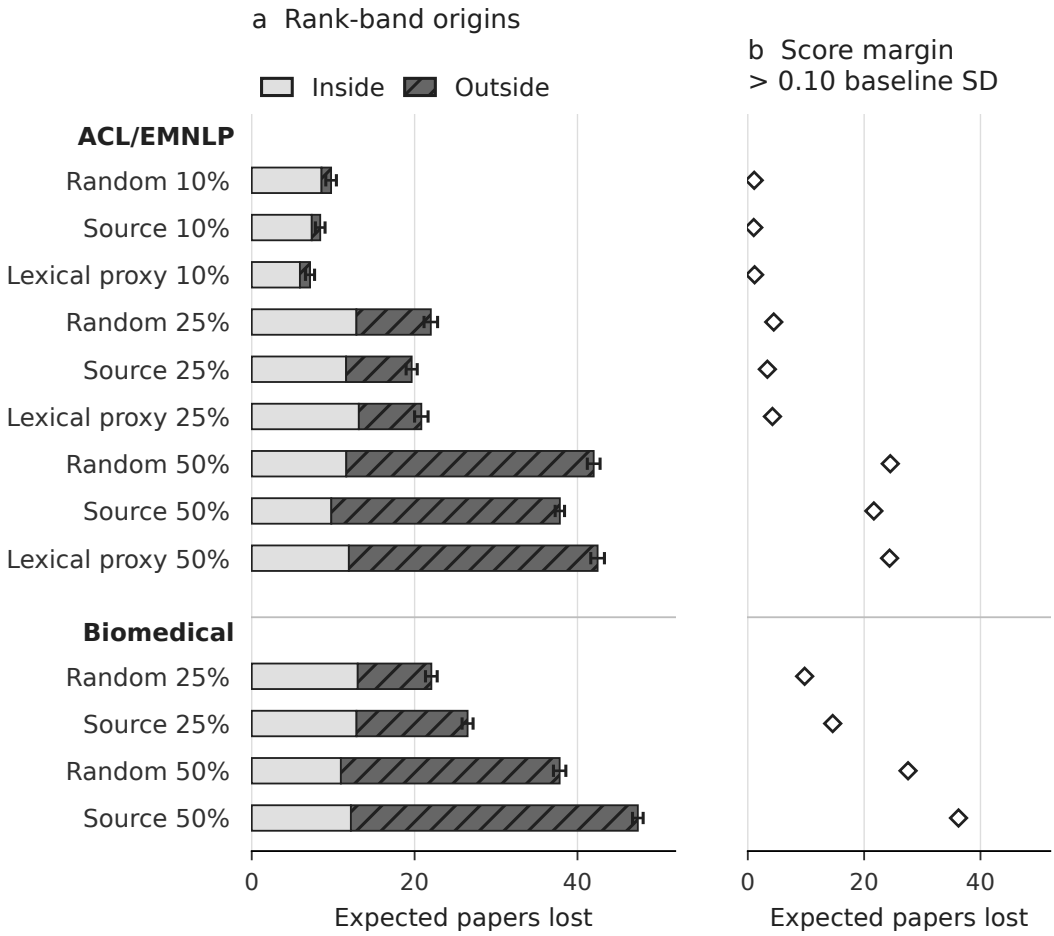


Figure S1: Original boundary-qualified mean loss counts across 200 assignments. Panel a partitions losses inside and outside the fixed rank band; original simultaneous total-loss intervals are retained only on totals. Panel b shows losses starting strictly above 0.10 baseline SD. The nested sets overlap and must not be added or treated as independent evidence. The band contains 16 baseline selections; the capacity bound in Section S13.2 qualifies outside-band counts. No intervals are assigned to partition components or panel b

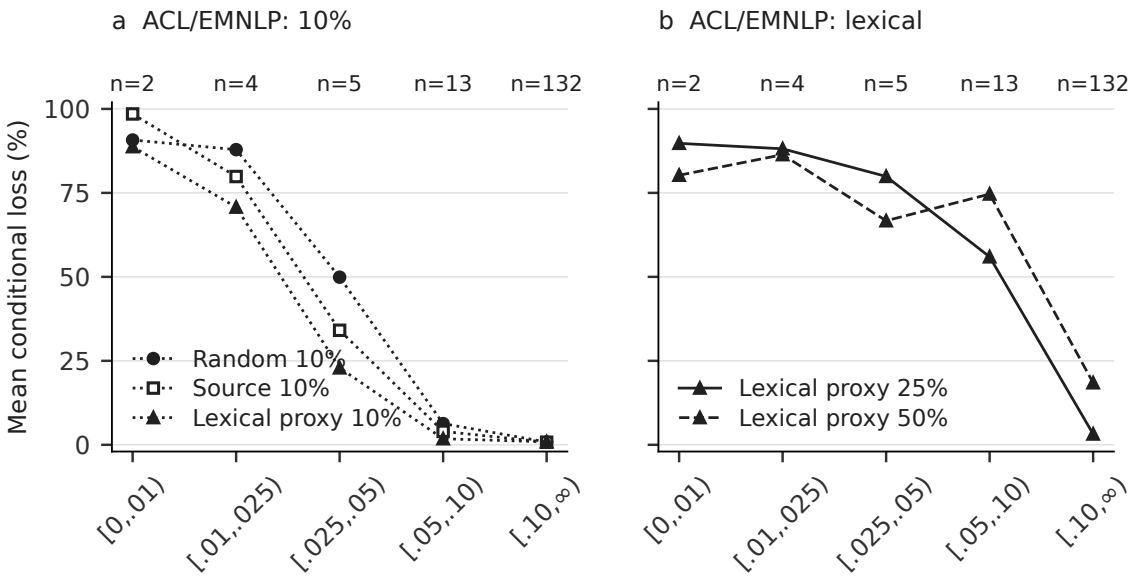


Figure S2: Remaining fixed ACL conditions: conditional loss among baseline-selected papers by signed score-margin bin. Panel a contains the three 10% conditions, and panel b lexical-preferred 25%/50%. Bin counts are fixed and shown above the axes. The categorical horizontal axis has unequal bin widths and an open-ended last bin. Curves are descriptive means across 200 assignments without new confidence intervals. Empty bins are undefined; these positive bins are nonempty